

You, Me, PuppetDB, and

Immutable Infrastructure

Code Mesh 2013

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Let's talk about...

Sorry

NO

CHANGE

**Immutability
is great!**

**Immutability allows
for invariants, which
help you reason about
correctness**

**Immutability prevents
spooky action at a
distance**

Immutability fosters
modular, composable
abstractions

*(not a tough sell
to functional
programmers)*

**That's great for
development, but how
about operations?**

Immutability for infrastructure?

*Because operations is in the
same boat as development*

Everyone who's got
their app running on
a fleet of servers has
experienced **spooky
action at a distance**

**Known, good state is
critical for reliable
upgrades**

**A lack of predictability
in your systems
ruins automation and
abstraction**

Chance

Poor invariants!
Lose \$172,222
per second





↓3.9695
-2.97↓-42.80%

VNTV
Vantiv Inc

↓54.7200
-0.27↓-0.49%

EV
Eaton Vance

↑12.5900
0.47↑3.88%

↑24.3800
-0.27↓-1.10%

WCC
Wesco International

↓6.2500
-0.86↓-12.10%

AB
Alliance Bernstein Holding LP

↑26.5950
0.01↑0.02%

7.0800
-0.28%

MWW
Monster Worldwide Inc

↓26.7150
0.01↑0.06%

PRI
Primerica Inc

↓20.0700
0.09↑0.45%

200
0.25%

LAZ
Lazard Ltd

↑4.7800
0.09↑1.92%

OLN
Olin

300
80%

GCAP
Gain Capital Holdings Inc

↓3.9695
-2.97↓-42.80%

3.96
5

3.99
HIGH

3.97
41

.70
17

KCG
Knight Capital Group Inc

42,178,624
VOLUME

3.27
OPEN

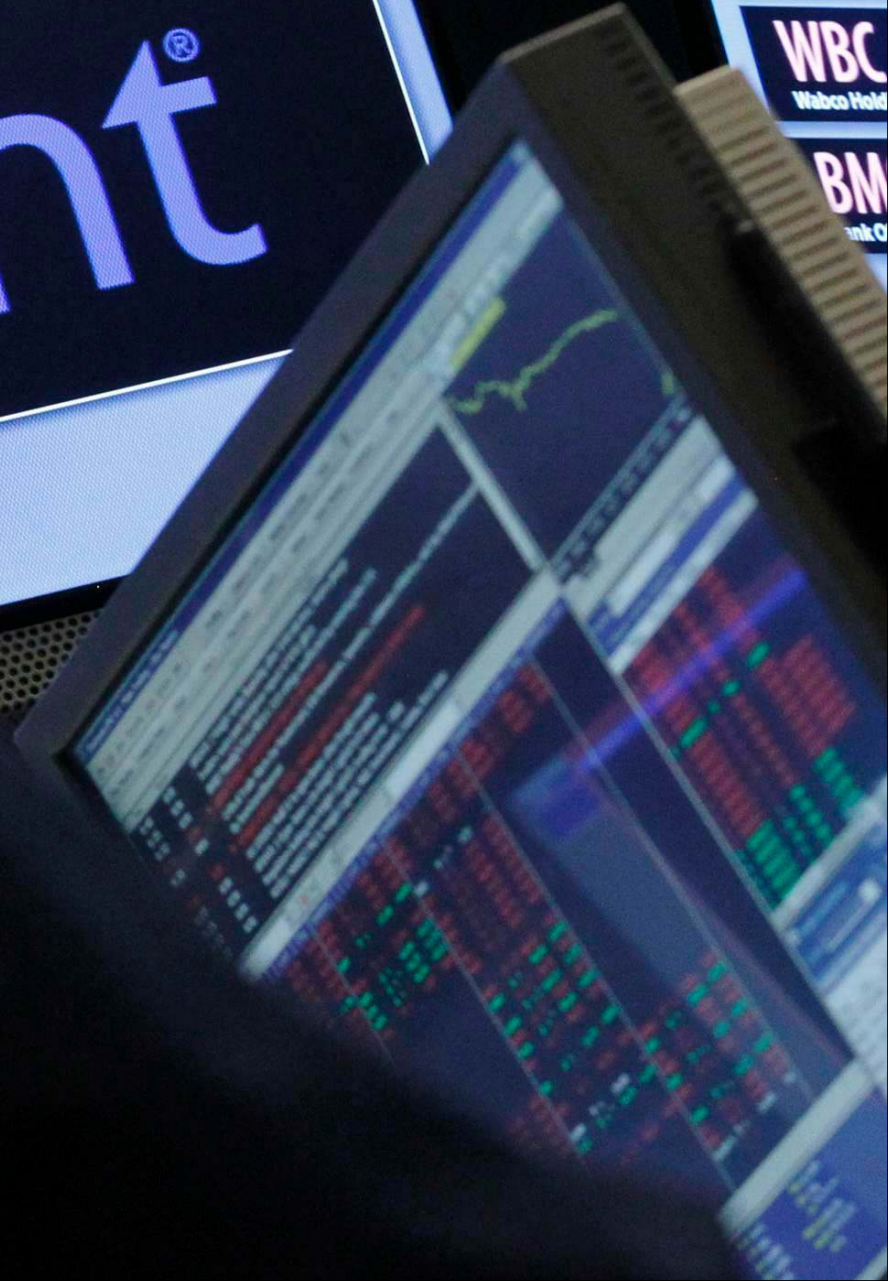
3.21
LOW

Knight®

1700

A

SECURITIES



The problem is that:

**Systems are inherently
mutable!**

But ideally:

**Systems should behave as
though they weren't!**

Computer systems are in many ways open systems, providing the keys to the vault if one is so inclined to grab them. But in order to foster an air of immutability in our own systems, it's of utmost importance to create a façade of immutability. Immutability requires that we layer over and abstract the parts of our system that provide unrestrained mutability.



Describe how you'd
like your systems to
look, and **Puppet does
all the hard work for
you!**

```
file { "/etc/issue":  
    content => "Got an issue? Here's a tissue!",  
}
```

```
file { "/etc/motd":  
    content => template("Welcome to $hostname!"),  
}
```

```
file { "/etc/sudoers":  
    owner    => root,  
    group    => root,  
    mode     => 440,  
    source   => "puppet:///modules/sudo/sudoers"  
}
```

```
package { 'ntp':  
  ensure => installed,  
}  
  
service { 'ntpd':  
  ensure      => running,  
  enable      => true,  
  subscribe => File['/etc/ntp.conf'],  
}  
  
file { '/etc/ntp.conf':  
  ensure => file,  
  require => Package['ntp'],  
  source  => "puppet:///modules/ntp/ntp.conf",  
}
```



```
class ntp {  
  
  package { 'ntp':  
    ensure => installed,  
  }  
  
  service { 'ntpd':  
    ensure      => running,  
    enable      => true,  
    subscribe => File['/etc/ntp.conf'],  
  }  
  
  file { '/etc/ntp.conf':  
    ensure => file,  
    require => Package['ntp'],  
    source => "puppet:///modules/ntp/ntp.conf",  
  }  
  
}
```

```
node "webserver.mydomain.com" {  
    include ntp  
}
```

```
node "appserver.mydomain.com" {  
    include ntp  
}
```

```
node "database.mydomain.com" {  
    include ntp  
}
```

```
class ssh {  
  
    @@sshkey { $hostname:  
        type => dsa,  
        key => $sshdsakey  
    }  
  
    Sshkey <<| |>>  
  
}
```

File “/tmp/foo/bar”

User “deepak”

Dir “/tmp/foo”

Dir “/tmp”

Dir “/tmp”

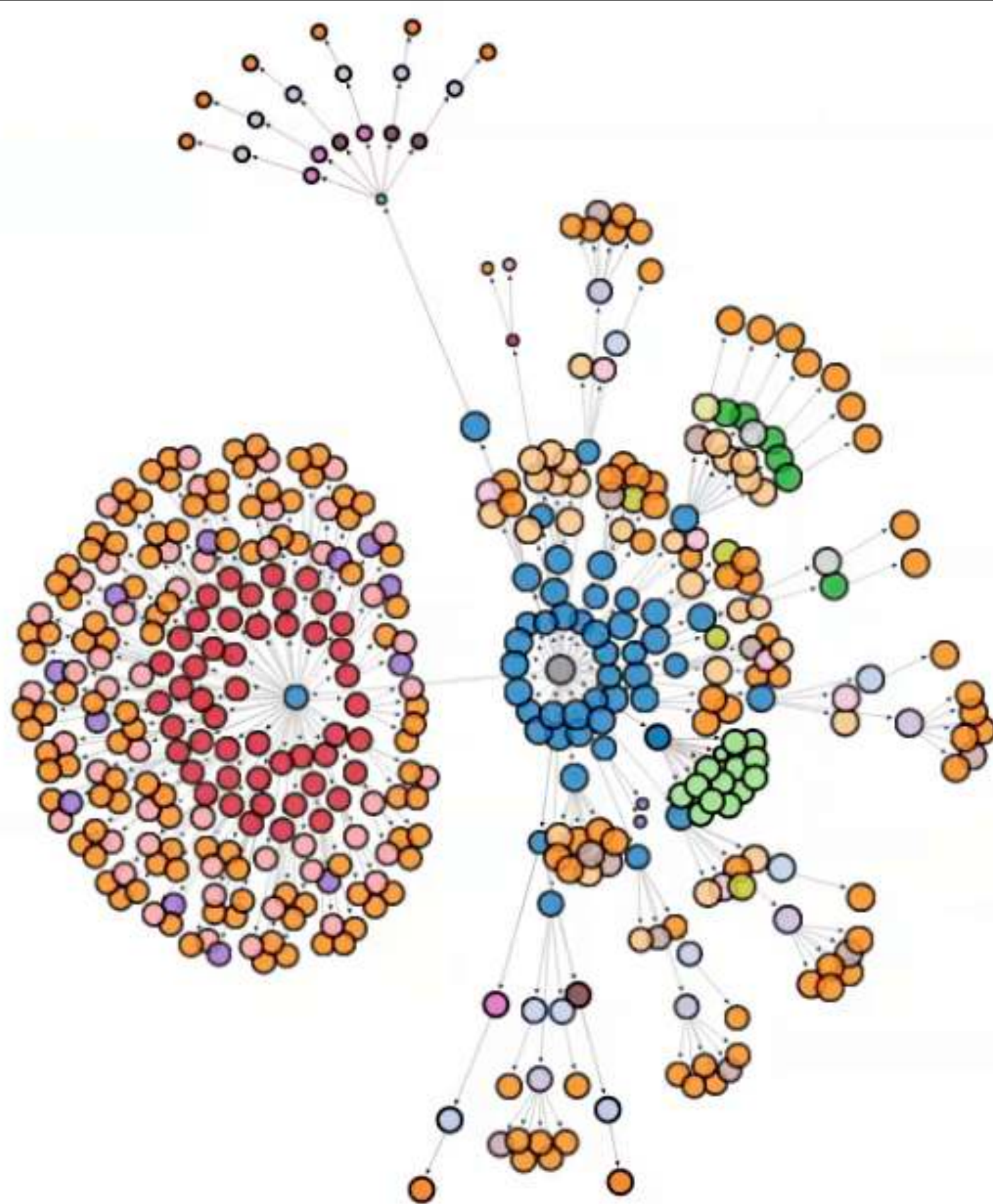
User “deepak”



Dir “/tmp/foo”



File “/tmp/foo/bar”



Idempotent, and only
does what's
necessary

Compensates for the
inherent mutability
of systems

**Combats spooky
action at a distance
with automatic
repair**

Brings **predictability**
to your *systems*

A foundation of
predictability and
reliability lets you
perform **higher-level
operations** on your
infrastructure

vOLUME

every resource
every parameter
every relationship
every fact
for every node
updated all the time

Users leverage this data to do **higher-order things** with their infrastructure

**key distribution
monitoring
clustered services
master/slave replication
load balancers
shared filesystems
firewall rules**

...

Infrastructure as code

Infrastructure as data

User demand:

Store as much data as we can!
Much better queryability!

Oh yeah, but:

Don't slow down the system!
Don't compromise reliability!

**We can rebuild it, we
have the technology!**

Speed is important

*Parsing, validating, and
manipulating incoming data is
computationally expensive*

Speed is important

The slower central storage is, the less agile sysadmins can be. That can cost money and uptime!

Reliability is
important

*If it's a critical part of managing
infrastructure, it's got to be solid!*

Deployment is important

*Should be easy to package, install,
configure, use, monitor, and
upgrade.*

**Wait, isn't Puppet
written in Ruby?**

Lessons learned from writing the rest of our software in Ruby

*It's...not speedy. Object creation,
method calls, garbage collection,
etc. “Magical” APIs amplify
the problem.*

Lessons learned from writing the rest of our software in Ruby

*Can only use one core!
Workarounds compromise
performance or simplicity*

Lessons learned from writing the rest of our software in Ruby

*Mutable state all over
the damn place!*

Lessons learned from writing the rest of our software in Ruby

Many struggles with the runtime. :(



-- Jeff Gagliardi



PuppetDB

*Definitely **B**etter!*

Fast, safe storage
of catalogs, facts, and
events

*like, *way* faster!*

HTTP APIs

for resource, fact,
node, report retrieval

*plenty of data, just
a “curl” away!*

Stop Raging
and
Start Living

Command Query Responsibility Separation

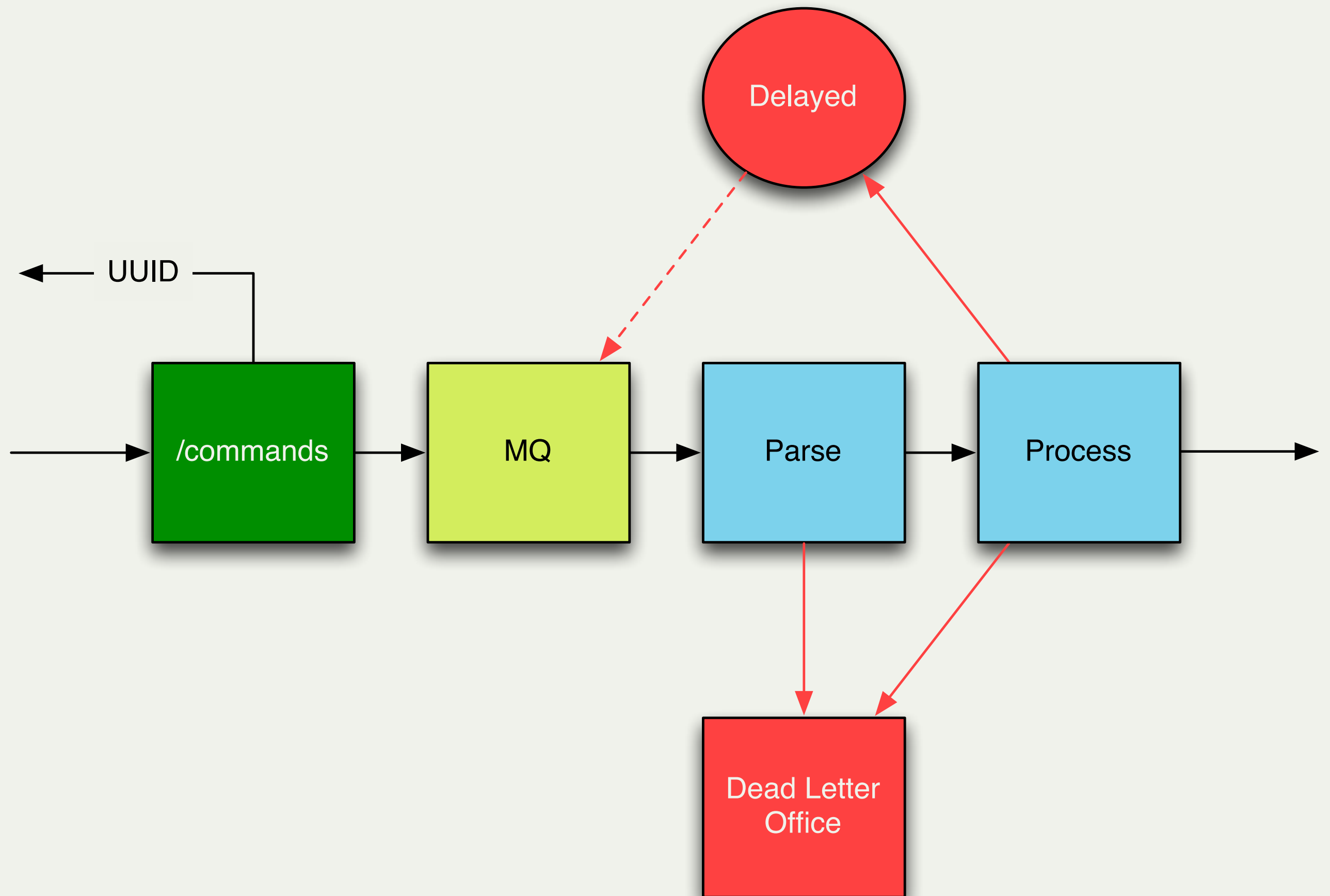
*use a different model to update
information than the model you
use to read information*

Writes

CORS-esque write pipeline

*async, parallel, MQ-based, with
automatic retry*

```
{  
  :command "replace catalog"  
  :version 2  
  :payload {...}  
}
```




```
(defmulti process-command!  
  (fn [{:keys [command version] :or {version 1}} _]  
    [command version]))  
  
(defmethod process-command! ["replace catalog" 1]  
  [command options]  
  (replace-catalog* command options))
```



```
(defmulti process-command!
  (fn [{:keys [command version] :or {version 1}} _]
    [command version]))

(defmethod process-command! ["replace catalog" 2]
  [command options]
  (replace-catalog* command options))

(defmethod process-command! ["replace catalog" 1]
  [command options]
  (-> command
    (update-in [:payload] catalog-v1->v2)
    (replace-catalog* options)))
```

Command processors
must be **retry-aware**

*expect failure, because
it ***will*** happen.*

(demo)

Reads

Queries

are expressed in their
own “language”

*domain specific, AST-based
query language*

```
["and",  
  ["=", "type", "User"],  
  ["=", "title", "nick"]]
```

```
["and",  
  ["=", ["fact", "operatingsystem"], "Debian"],  
  ["<", ["fact", "uptime_seconds"], 10000]]
```

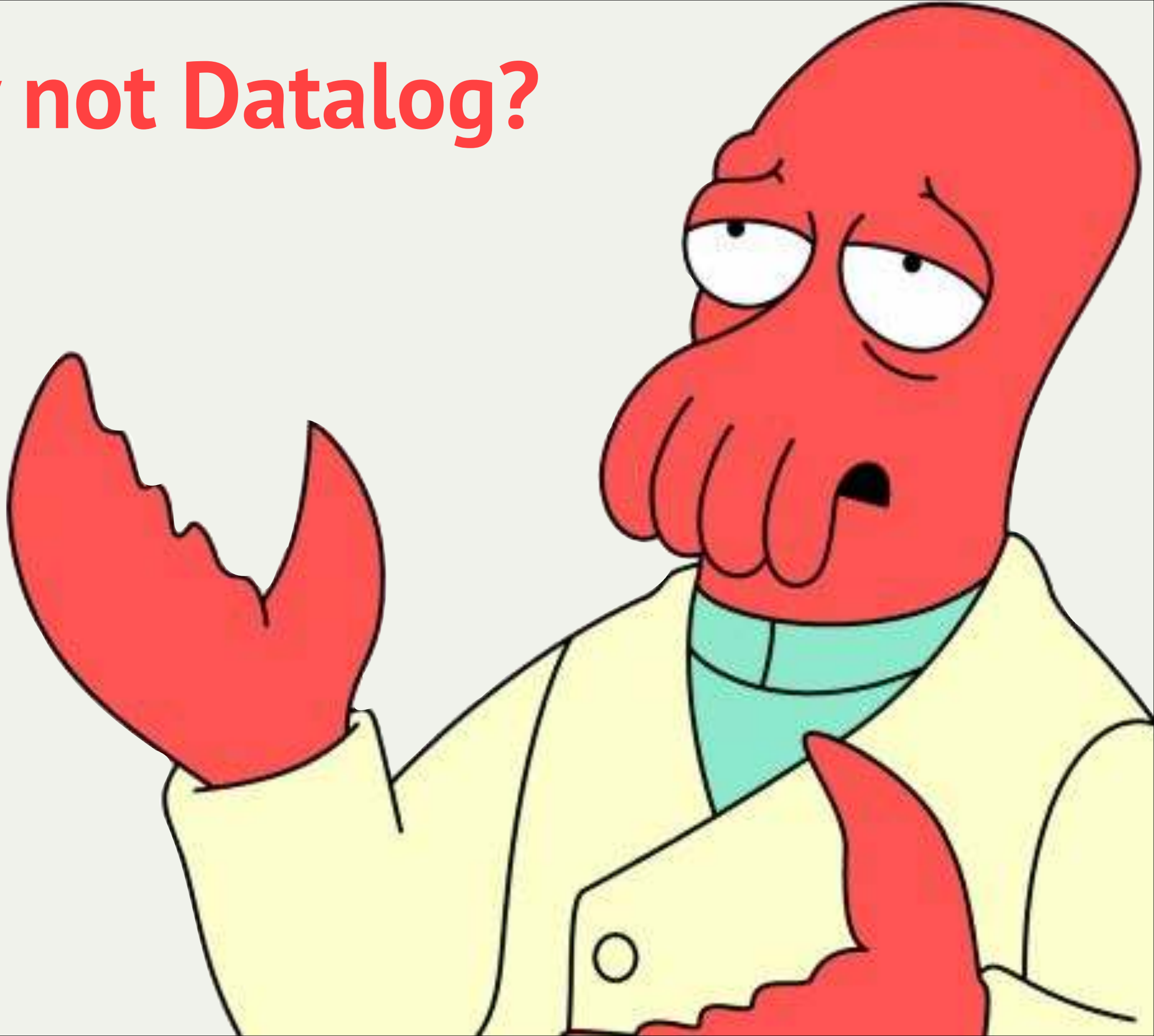


```
[ "and",
  ["=", "name", "ipaddress"],
  ["in", "certname",
    ["extract", "certname", ["select-resources",
      ["and",
        ["=", "type", "Class"],
        ["=", "title", "Apache"]]]]]]
```



```
["or",  
  ["=", "certname", "foo.com"],  
  ["=", "certname", "bar.com"],  
  ["=", "certname", "baz.com"]]
```

Why not Datalog?



We use `core.match`
to walk the tree,
compiling it to SQL

AST-based API lets
users **write their own**
languages and
manipulate queries

*ah, you've got to love
open source!*

```
(Package[httpd] and country=fr)  
or country=us
```

```
Package["mysql-server"]  
and architecture=amd64
```

Erik Dalén, Spotify

<https://github.com/dalen/puppet-puppetdbquery>

The screenshot shows the Puppetboard web interface in a browser window. The address bar shows the URL `0.0.0.0:5000/node/puppetdb.nedap.local`. The navigation bar includes links for PUPPETBOARD, OVERVIEW, NODES, FACTS, REPORTS, and QUERY. The main content area is divided into three sections: Details, Facts, and Reports.

Details

Hostname	puppetdb.nedap.local
Catalog compiled at	Tuesday, August 6 2013 3:00 PM
Facts retrieved at	Tuesday, August 6 2013 3:00 PM
Report uploaded at	Tuesday, August 6 2013 3:00 PM

Facts

Type here to filter

Fact	Value
architecture	amd64
augeasversion	0.10.0
bios_release_date	12/01/2006
bios_vendor	innotek GmbH
bios_version	VirtualBox
blockdevice_sda_model	VBOX HARDDISK
blockdevice_sda_size	16106127360
blockdevice_sda_vendor	ATA
blockdevice_sr0_model	CD-ROM
blockdevice_sr0_size	1073741312
blockdevice_sr0_vendor	VBOX
blockdevices	sda,sr0
boardmanufacturer	Oracle Corporation
boardproductname	VirtualBox
boardserialnumber	0
clientcert	puppetdb.nedap.local
clientversion	3.2.2
domain	nedap.local
facterversion	1.7.1

Reports

Only showing the last ten reports.

Start time	Run time	Full report
Tuesday, August 6 2013 3:00 PM	0:00:01	d7334a...

Copyright © 2013 [Daniele Sluijters](#). Live from PuppetDB.

daenny, Puppetboard

<https://github.com/nedap/puppetboard>

Chrome File Edit View History Bookmarks Window Help Tue 6:41 PM Wagner

Puppet Enterprise Event Inspector

localhost:9292/#

puppet enterprise console

node requests (0) admin Help

Events Nodes Groups Classes Reports Inventory Search Live Management

Summary / Resources with failures / Exec[10gen-apt-update]

resources: with failures

2 resources that had events and the number of nodes affected:

Resource
Exec[10gen-apt-update]
Package[mongodb-10gen]

resource: Exec[10gen-apt-update]

49 events for this resource and the nodes affected:

3 Failed | 46 Changed

Event	Node	Class	Time
failure	dhcp69.redhat.backline.puppet...	Mongodb	2013-08-10 23:19:20
failure	dhcp71.redhat.backline.puppet...	Mongodb	2013-08-10 23:19:58
failure	dhcp80.redhat.backline.puppet...	Mongodb	2013-08-10 23:20:06
success	dhcp62.redhat.backline.puppet...	Mongodb	2013-08-10 23:55:03
success	dhcp55.redhat.backline.puppet...	Mongodb	2013-08-10 23:55:07
success	dhcp74.debian.backline.puppe...	Mongodb	2013-08-10 23:17:57
success	dhcp73.debian.backline.puppe...	Mongodb	2013-08-10 23:17:59
success	dhcp65.debian.backline.puppe...	Mongodb	2013-08-10 23:17:55
success	dhcp89.debian.backline.puppe...	Mongodb	2013-08-10 23:18:55
success	dhcp64.debian.backline.puppe...	Mongodb	2013-08-10 23:18:28
success	dhcp86.debian.backline.puppe...	Mongodb	2013-08-10 23:19:07
success	dhcp51.debian.backline.puppe...	Mongodb	2013-08-10 23:19:12
success	dhcp57.debian.backline.puppe...	Mongodb	2013-08-10 23:18:37
success	dhcp98.debian.backline.puppe...	Mongodb	2013-08-10 23:18:45
success	dhcp52.debian.backline.puppe...	Mongodb	2013-08-10 23:19:43
success	dhcp67.debian.backline.puppe...	Mongodb	2013-08-10 23:19:54

Puppet Enterprise, Event Inspector

<https://puppetlabs.com>

Foreman Integration (CERN)

https://github.com/cernops/puppetdb_foreman

Web UI

<https://github.com/dima-exe/puppetdb-db>

Web UI

<https://github.com/gbougeard/puppetdb-frontend>

Ruby

<https://github.com/dalen/puppet-puppetdbquery>

Ruby (DataMapper)

<https://github.com/dalen/dm-puppetdb-adapter>

Ruby

<https://github.com/ripienaar/ruby-puppetdb>

Python

<https://github.com/nedap/pypuppetdb>

Python

<https://github.com/arcus-io/puppetdb-python>

Python

<https://github.com/JHaals/puppetdb-grep>

Java

<https://github.com/thallgren/puppetdb-javaclient>

Go

<https://github.com/nightlyone/puppetquery>

Scala

<https://github.com/gbougear/puppetdb-frontend>

CoffeeScript

<https://gist.github.com/pmuellr/5591686>

Node.js

<https://github.com/nightfly19/minidb>

MCollective

<https://github.com/ploubser/mcollective-puppetdb-discovery>

Rundeck

<https://github.com/sirhopcount/puppetdb-rundeck>

Rundeck

<https://github.com/martin2110/puppetdb-rundeck>

OpenStack

https://github.com/bodepd/puppet-openstack_puppetdb

Vagrant

<https://github.com/grahamgilbert/vagrant-puppetmaster>

PowerDNS

<https://github.com/evenup/evenup-pdns>

```
(def query-app
  (app
    [&]
    { :get (fn [{:keys [params] :as request}]
              (perform-query (params "query"))) }) ) )
```

```
(def resources-app
  (app
    []
    query-app

    [type title &]
    (comp query-app
      (partial http-q/restrict-resource-query-to-type type)
      (partial http-q/restrict-resource-query-to-title title))

    [type &]
    (comp query-app
      (partial http-q/restrict-resource-query-to-type type)))) )
```

```
(defn restrict-resource-query-to-type
  [type query]
  (conj ["and"
        ["=" "type" type]]
        query) )
```


Storage

Relational Database, embedded or PostgreSQL

*because they're actually pretty
fantastic at ad-hoc queries,
aggregation, windowing, etc.
while maintaining safety*

Fault-tolerance

by @jrecursive



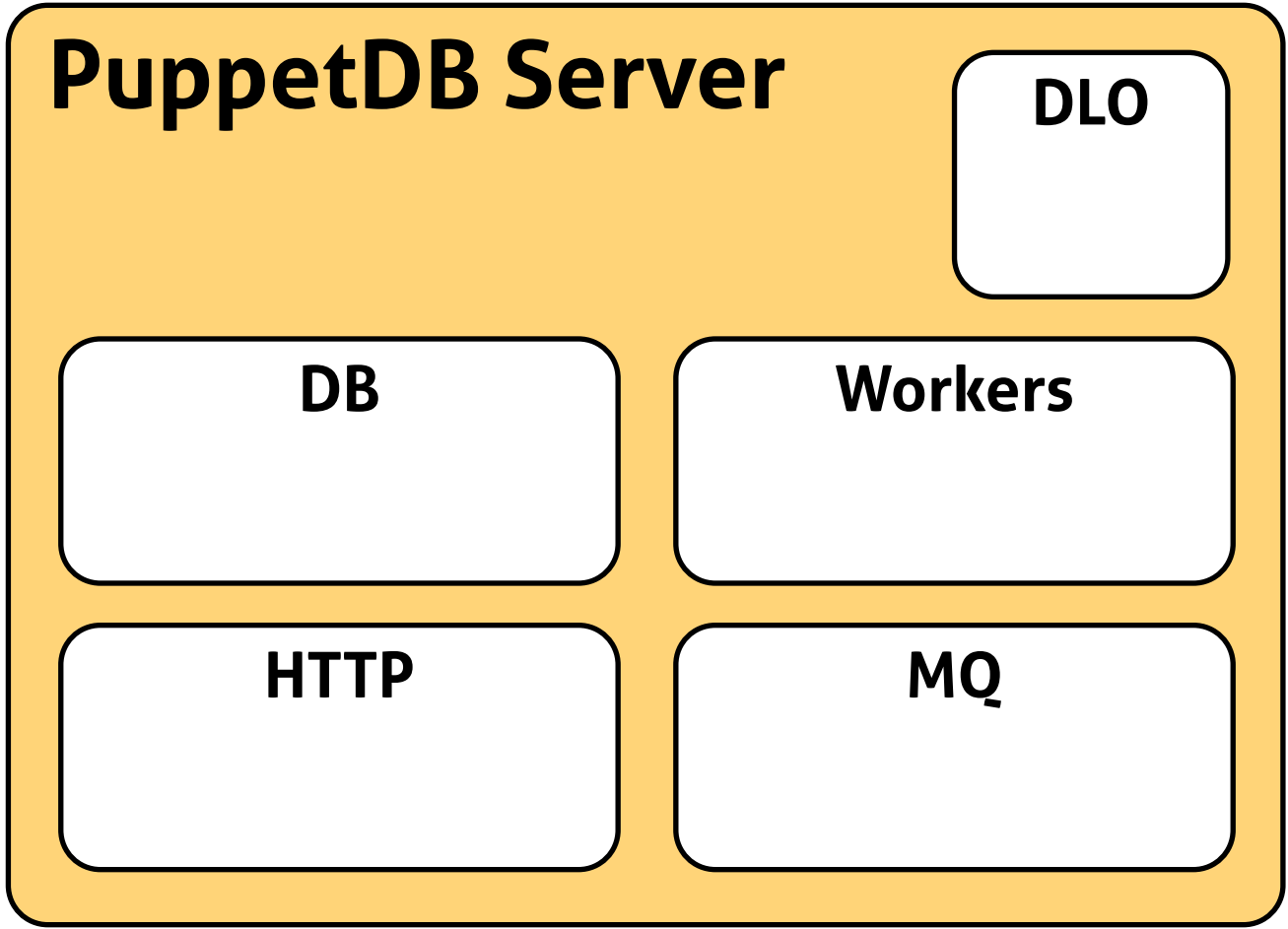
Relational Database, embedded or PostgreSQL

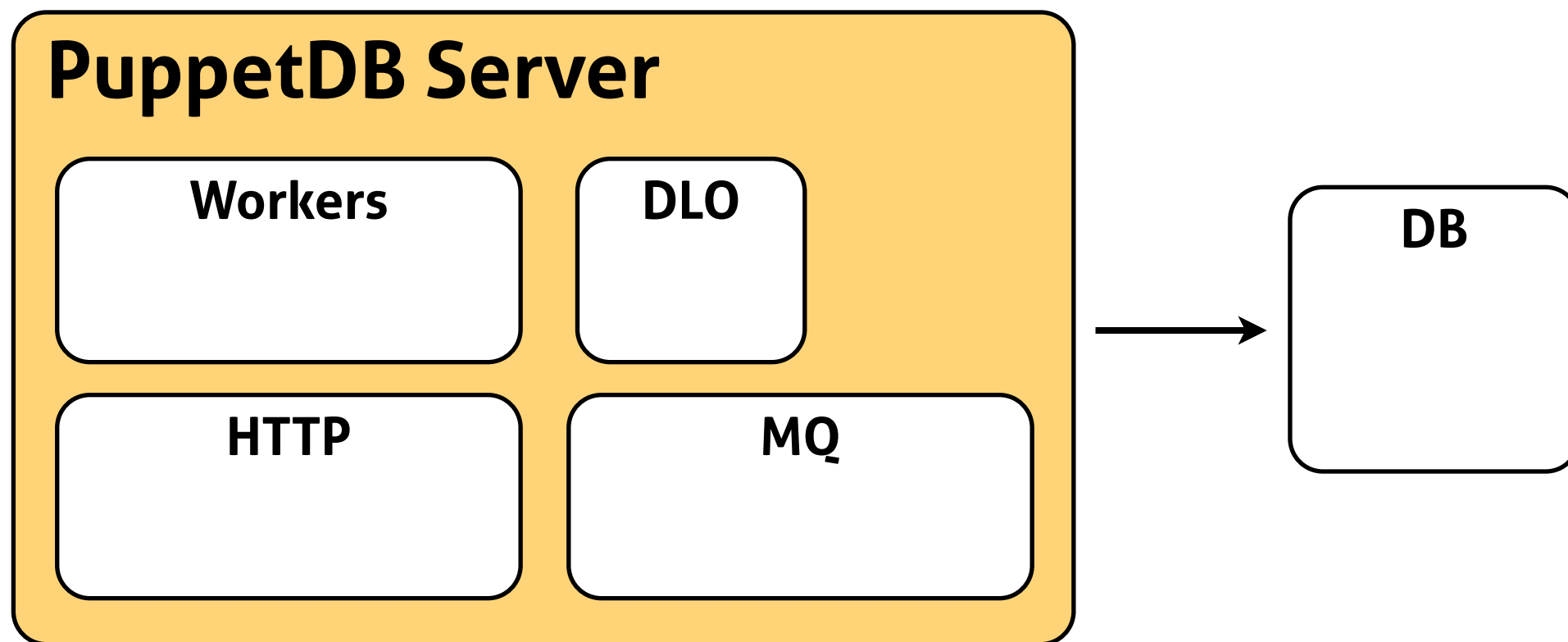
*we use arrays, recursive queries,
indexing inside complex
structures*

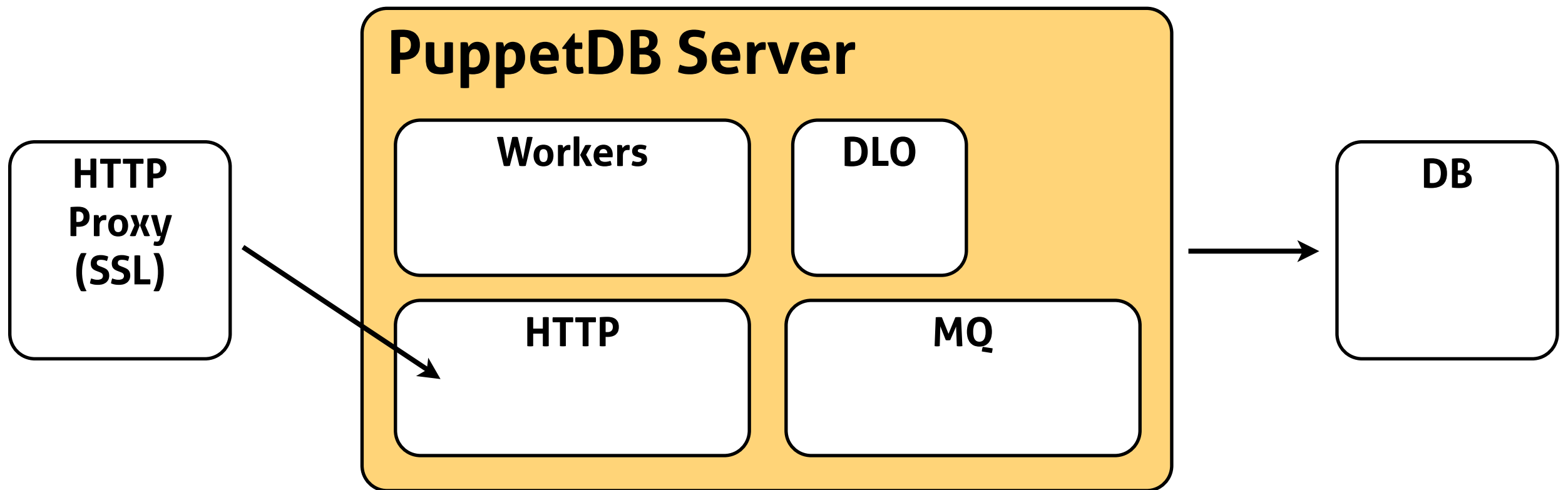
Relational Database, embedded or PostgreSQL

*schema is oriented towards the
types of queries we encounter*

Deployment







Results

Thousands of production deployments

Small shops with a dozen hosts,
large shops with thousands of
hosts, intercontinental
deployments...

There is a new
deployment of
PuppetDB **every**
15 minutes.



Dean Wilson @unixdaemon

7h

Did some basic performance testing of **#puppetdb** and reduced our compile times from 180ish second to mid 20s. **#massivewin**

Expand



Greg Mason @greg_mason

25 Sep

according to **#puppetdb**, I'm currently managing 71,777 resources with Puppet.

Expand



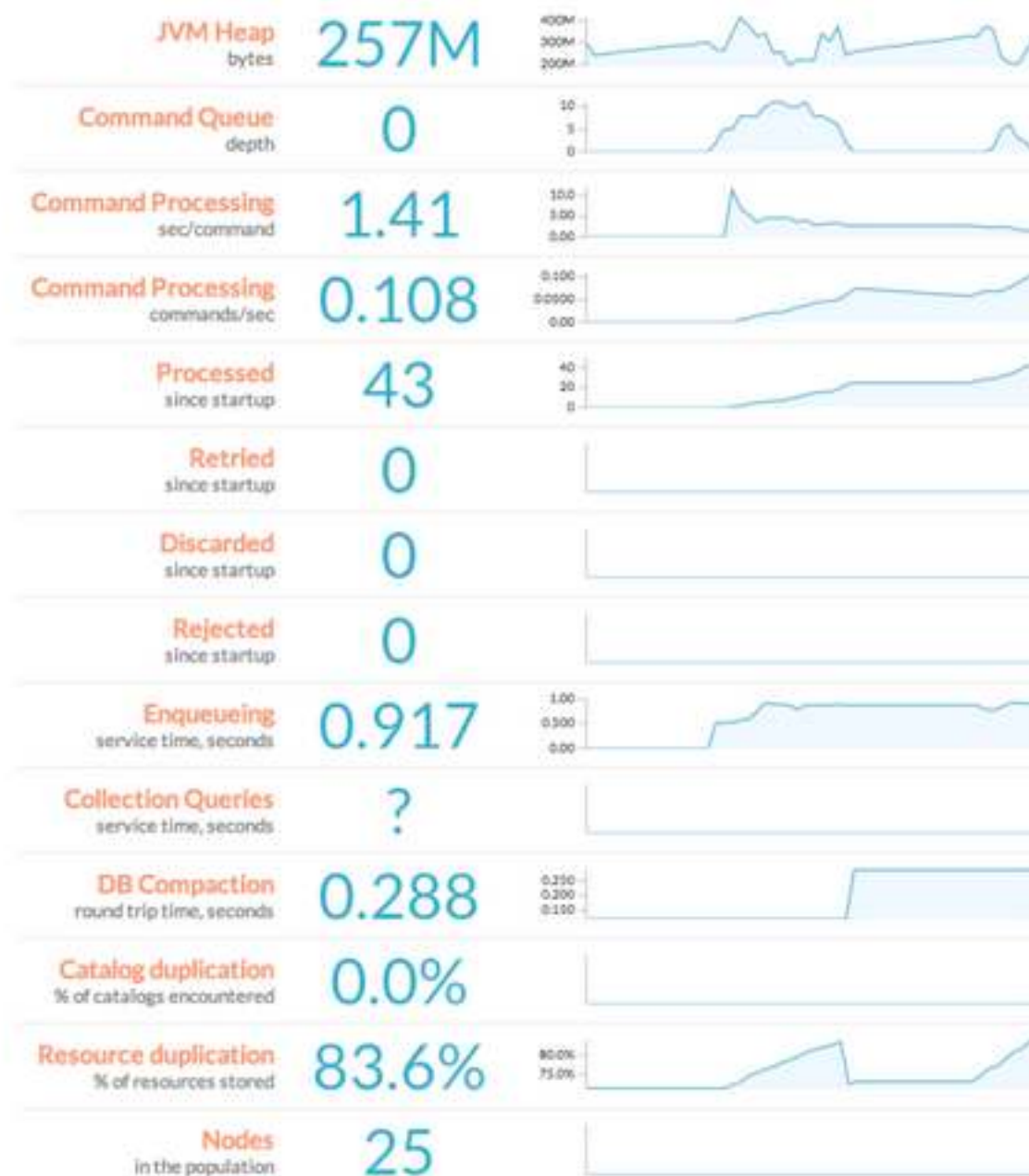
ripienaar @ripienaar

13 Jun

puppetdb + 25 masterless nodes with embedded database
[twitpic.com/9vz7y2](https://twitter.com/ripienaar/status/9vz7y2)

Retweeted by D Giridharagopal

Hide photo Reply Retweeted Favorite

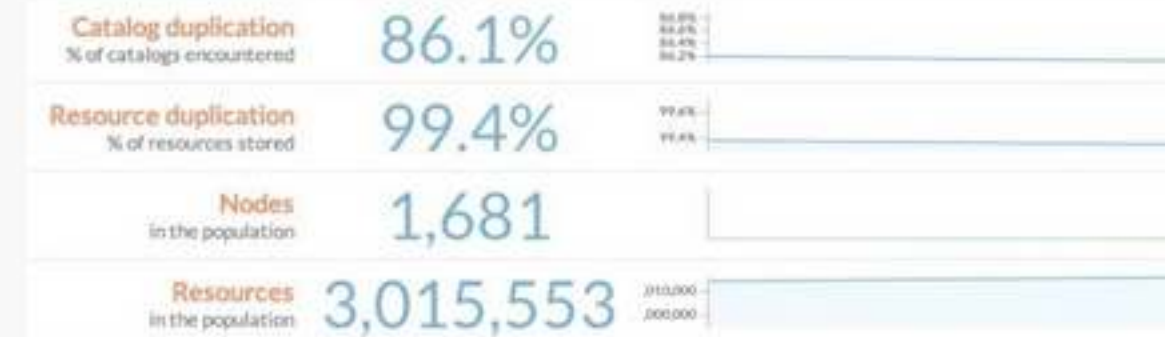


Matt Nicholson @sjoebou

20 Jul

@glarizza @grim_radical two pic.twitter.com/mvJuOiJF

Hide photo



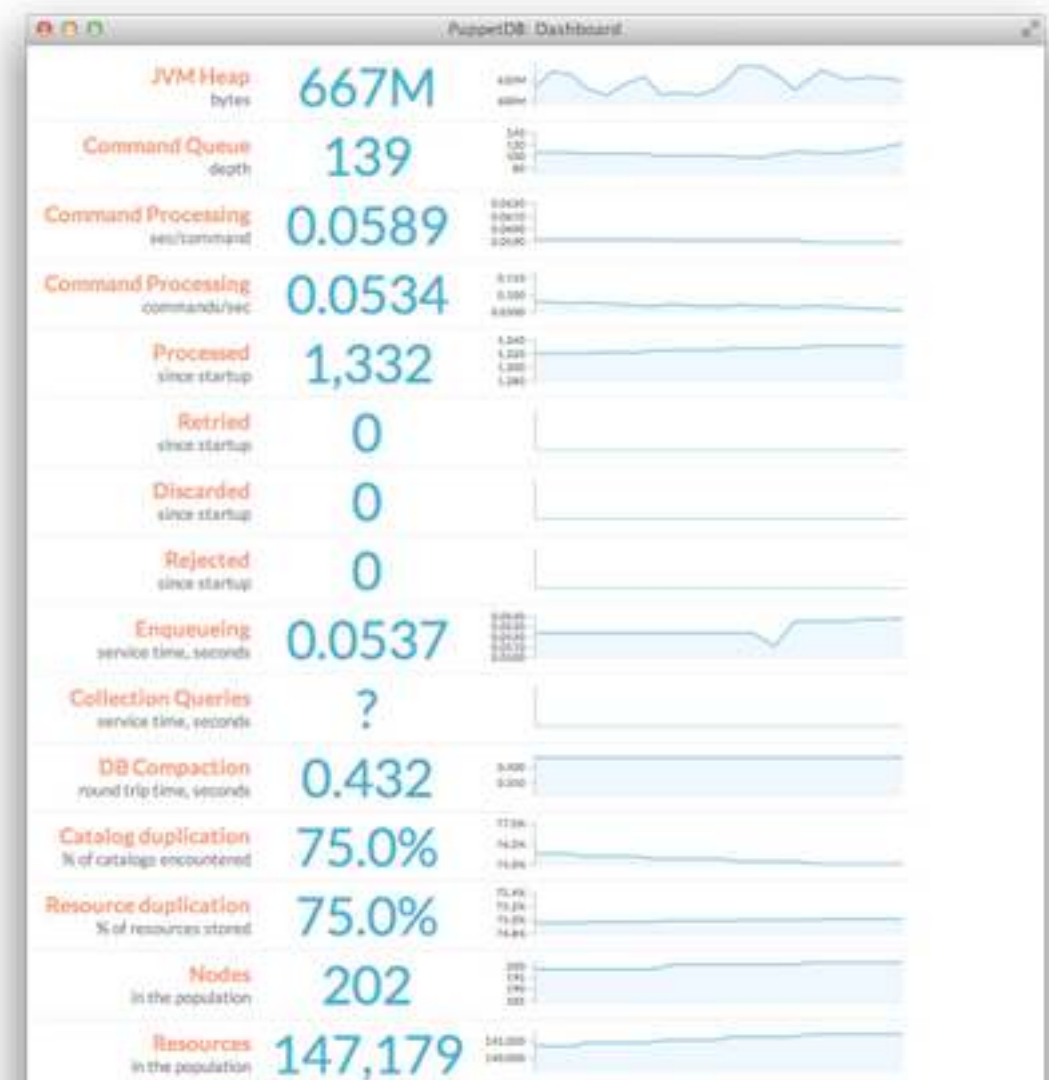
Lars Fronius @LarsFronius

23 May

This **#puppetdb** dashboard thingy is actually so awesome. Oh, and puppetdb is such a big step in performance! **#puppetize**
[twitpic.com/9oi290](https://twitter.com/LarsFronius/status/9oi290)

Retweeted by D Giridharagopal

Hide photo Reply Retweeted Favorite



Thousands of deployments,
Lots of threads per install,
Zero deadlocks,
Zero bugs involving clojure state

*companion Ruby code has
~10x the defect rate*

Users dig the uberjar,
makes deployment
straightforward and
less error-prone

**Nobody cares that it's
Clojure.** Users just
want stuff to work.

Monitoring and instrumentation is a big deal. Users want easy ways to consume metrics.

Nagios

<https://github.com/jasonhancock/nagios-puppetdb>

Nagios

<https://github.com/favoretti/puppetdb-external-naginator>

Munin

https://github.com/vpetersson/munin_puppetdb

Munin

<https://github.com/dalen/puppetdb-muninplugins>

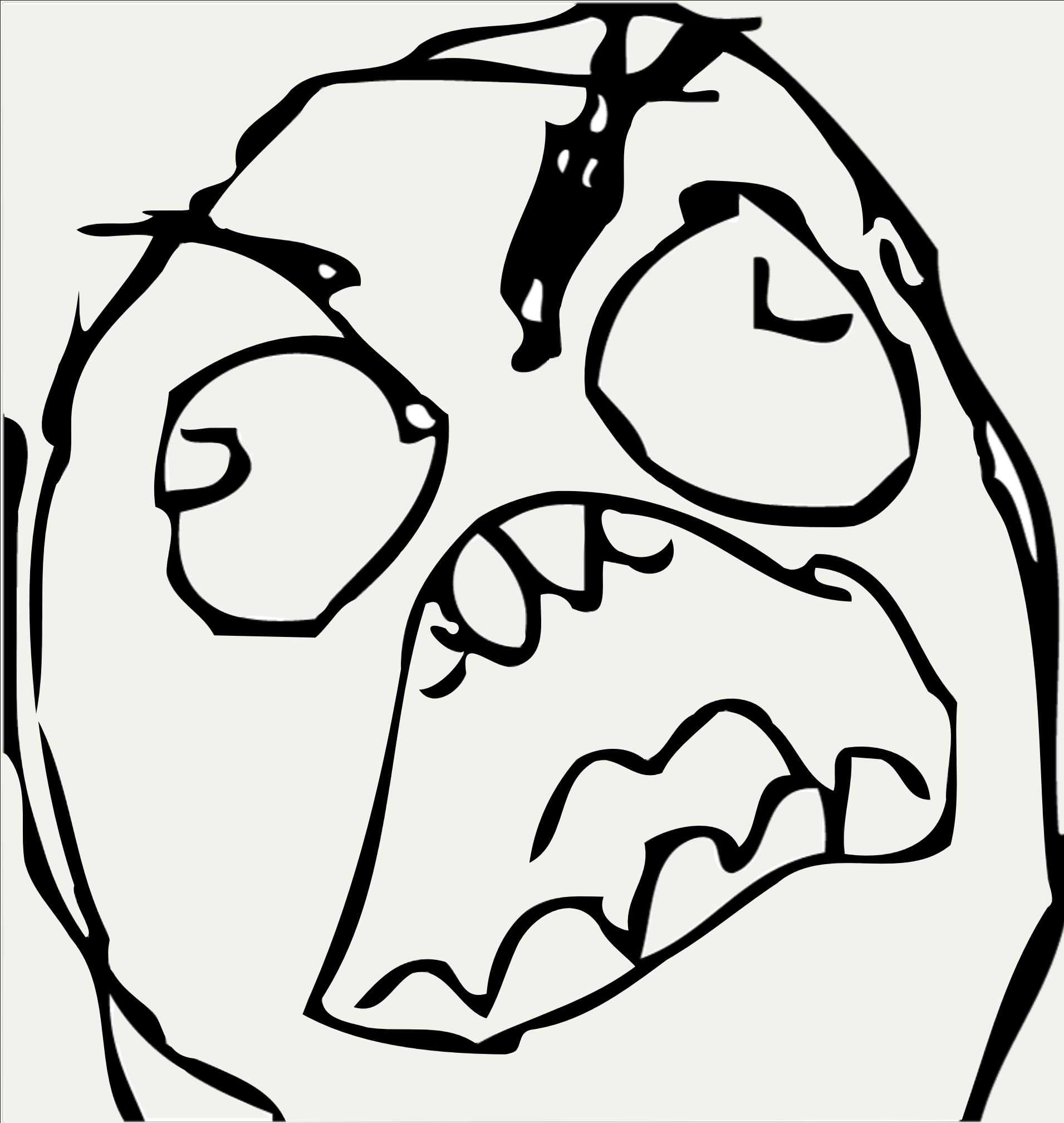
Collectd

<https://gist.github.com/mfournier/5615125>

Open source

[http://github.com/
puppetlabs/puppetdb](http://github.com/puppetlabs/puppetdb)

**If you build it, they
will come.**



FFFFFFFF

FFFFFFFF

FFFFFFF

FFFUU

UUUU

UUUU

UUUU

UUUU

UUUU-

**Think about the
entire stack**

I don't care what tools
you use, but *please use*
good ones

deepak
giridharagopal

deepak@puppetlabs.com

@grim_radical [github twitter freenode]

We're hiring!